

MULTIMODAL MISSENSE VARIANT PATHOGENICITY PREDICTION

Comprehensive Combinatorial Configuration Matrix, Feature Taxonomy & Architecture Space

Combinatorial Space: >10⁵ Paths

10 Geom Ablations: 1,023 Subsets

Strict Metric: MCC

1. SEQUENCE CONTEXT SPACE

5 x 5 = 25 Contexts

[DNA] Context Lengths (Centered on SNV):

301 bp	: Baseline local flanking genomic sequence
1001 bp	: Proximal promoter & exon-intron boundaries
2001 bp	: [*] Optimal genomic context window
5001 bp	: Extended cis-regulatory / enhancer regions
10001 bp	: Long-range chromatin interaction domain

[Protein] Context Lengths (Centered on Missense):

31 aa	: Immediate local peptide microenvironment
101 aa	: Secondary structure & conserved motif window
251 aa	: Domain-scale functional folding context
501 aa	: Multi-domain functional interaction context
1001 aa	: [*] Grand optimal full-length protein context

2. FOUNDATION MODELS

3 x 4 = 12 Model Pairs

[DNA Models] 3 Genomic Foundation Models:

1.1 nt_v1_500m	MLM, Human reference genome
1.2 nt_v2_500m	[*] Multi-species SOTA DNA model
1.3 nt_v3_650m	MLM, Next-Gen Transformer Architecture

[Protein Models] 4 Protein Language Models (pLM):

2.1 esm1b_650m	[*] Classic BERT pLM (UniRef50 Benchmark)
2.2 esm1v_650m	Zero-shot variant effect 5-model ensemble
2.3 esm2_650m	Modern RoPE SOTA Evolutionary Scale Model
2.4 esmc_600m	Next-Gen ESM Cambrian Model (Autoregressive)

3. POOLING & EMBEDDINGS

3 Pooling x 2 Embeds

[Pooling] 3 Token Aggregation Strategies:

* Mean Pooling	-> Non-pad token average; captures broad global sequence context
* Center Pooling	-> Direct token extraction at the exact missense mutation site
* CLS Pooling	-> Sequence-level representation from leading [CLS]/Header token

[Embeddings] Continuous Latent Shift Representations:

* DNA Embed	: Nucleotide latent space displacement
* Prot Embed	: Amino acid evolutionary space displacement
* Delta Vector	: Preserves mutation directionality (Latent Shift)

4. BIOLOGICAL FEATURES

2 AF + 9 CS (11 Total)

[AF] 2 Allele Frequency Features (Population Genetics):

1.1 AF	: 1000 Genomes Project global alternate allele frequency
1.2 gnomADe_AF	: gnomAD Exome sequencing global population allele frequency

[CS] 9 Evolutionary Conservation Scores:

2.1 phyloP100way_vertebrate	: PhyloP conservation score across 100 vertebrate genomes
2.2 phyloP470way_mammalian	: PhyloP conservation score across 470 mammalian genomes
2.3 phyloP17way_primate	: PhyloP conservation score across 17 primate genomes
2.4 phastCons100way_vertebrate	: PhastCons conserved element probability across 100 vertebrates
2.5 phastCons470way_mammalian	: PhastCons conserved element probability across 470 mammals
2.6 phastCons17way_primate	: PhastCons conserved element probability across 17 primates
2.7 GERP++_RS	: GERP++ Rejected Substitutions (evolutionary constraint score)
2.8 GERP++_NR	: GERP++ Neutral Rate (expected neutral substitution rate)
2.9 GERP_92_mammals	: Standardized GERP score aligned across 92 mammalian species

5. SPLICEAI SCORES

9 Disruption Scores

[DS] 4 Delta Probability Scores:

1.1 SpliceAI_pred_DS_AG	: Delta Score - Acceptor Gain probability (novel acceptor site)
1.2 SpliceAI_pred_DS_AL	: Delta Score - Acceptor Loss probability (loss of native site)
1.3 SpliceAI_pred_DS_DG	: Delta Score - Donor Gain probability (novel donor splice site)
1.4 SpliceAI_pred_DS_DL	: Delta Score - Donor Loss probability (loss of native donor site)

[DP & Max] 4 Delta Positions & 1 Summary Score:

2.1 SpliceAI_pred_DP_AG	: Delta Position - Acceptor Gain distance (bp to novel site)
2.2 SpliceAI_pred_DP_AL	: Delta Position - Acceptor Loss distance (bp to lost native site)
2.3 SpliceAI_pred_DP_DG	: Delta Position - Donor Gain distance (bp to novel donor site)
2.4 SpliceAI_pred_DP_DL	: Delta Position - Donor Loss distance (bp to lost donor site)
2.5 SpliceAI_pred_DS_max	: [*] Max Delta Score = max(DS_AG, AL, DG, DL) [Splicing impact]

6. GEOMETRIC FEATURES

10 Features (1,023 Sets)

[DNA Geom] 5 DNA Manifold Descriptors:

1.1 dna_LLRL	: Log-Likelihood Ratio on DNA language model output distribution
1.2 dna_LVD_L2	: Euclidean L2 distance between Ref and Alt latent vectors
1.3 dna_LVD_Cosine	: Cosine angular deviation between Ref and Alt latent vectors
1.4 dna_LID	: Local Intrinsic Dimensionality (local manifold complexity k-NN)
1.5 dna_LVD_Relative	: Relative latent displacement normalized by baseline norm (V2)

[Protein Geom] 5 Protein Manifold Descriptors:

2.1 prot_LLRL	: [*] Top single predictor (+0.0171 MCC; >75% frequency in Top-20)
2.2 prot_LVD_L2	: Euclidean L2 distance in protein evolutionary embedding space
2.3 prot_LVD_Cosine	: Cosine angular displacement in pLM latent vector space
2.4 prot_LID	: Local Intrinsic Dimensionality k-NN on protein manifold
2.5 prot_LVD_Relative	: Relative latent displacement ratio of variant site (V2)

7. MULTIMODAL FUSION & CLASSIFICATION ARCHITECTURE SPACE (9 ARCHITECTURES)

3 Framework Families

Family 1: PyTorch End-to-End (4 Models)

End-to-end backpropagation training through specialized neural fusion modules

1. PyTorch_Concat

Linear Concatenation + Deep MLP Head

Direct concatenation of normalized multimodal feature vectors through deep feed-forward layers with BatchNorm & Dropout.

2. PyTorch_CrossAttn

Bidirectional Cross-Modal Attention

Dual cross-attention mechanism allowing DNA and Protein modalities to query mutual contextual representations.

3. PyTorch_Transformer

Multi-Head Self-Attention Fusion

Treats each modality as an independent context token, processed through Transformer Encoders for higher-order interactions.

4. PyTorch_Gating

Gated Multimodal Unit (Dynamic GMU)

Adaptive gating mechanism learning instance-dependent dynamic routing weights (w_dna, w_prot) across modalities.

Family 2: Pure Tree-Based (Top 1 SOTA)

Gradient Boosted Decision Trees directly on unified feature representations

[*] 5. Pure_XGBoost_Concat

PCA (k=32..64) + Tabular Concatenation + XGBoost

- Rank #1 Absolute Winner across all 4 benchmark datasets

- Peak Mean Performance: MCC = 0.6798 ~ 0.6995

Core Technical Advantages:

- Natural non-linear thresholding across Bio features (AF, GERP) and high-dimensional Foundation embeddings.
- Complete immunity to overfitting via PCA dimensionality reduction on 1280-dim latent representations.
- Optimal decision boundaries and superior sensitivity under severe pathogenic class imbalance.

Family 3: Hybrid Deep-Tree (4 Models)

Extract deep latent representation (f_global) from PyTorch to train an XGBoost Head

6. Hybrid_Concat_XGBoost

PyTorch Concat Embedding -> XGBoost

Trains PyTorch Concat network to extract high-level f_global vector, followed by XGBoost classification.

7. Hybrid_CrossAttn_XGBoost

Cross-Attention Latent Code -> XGBoost

Distills cross-modal interaction representations from Cross-Attention into boosted tree ensembles.

8. Hybrid_Transformer_XGBoost

Transformer Context Vector -> XGBoost

Extracts global context vectors from Self-Attention Encoders for gradient-boosted decision boundary learning.

9. Hybrid_Gating_XGBoost

Gated Weighted Representation -> XGBoost

Passes dynamically gated multimodal representations from GMU into an XGBoost classifier head.